

INNOPOLIS UNIVERSITY · BACHELOR'S THESIS · 2026

Secure Communication for Swarm Robotics

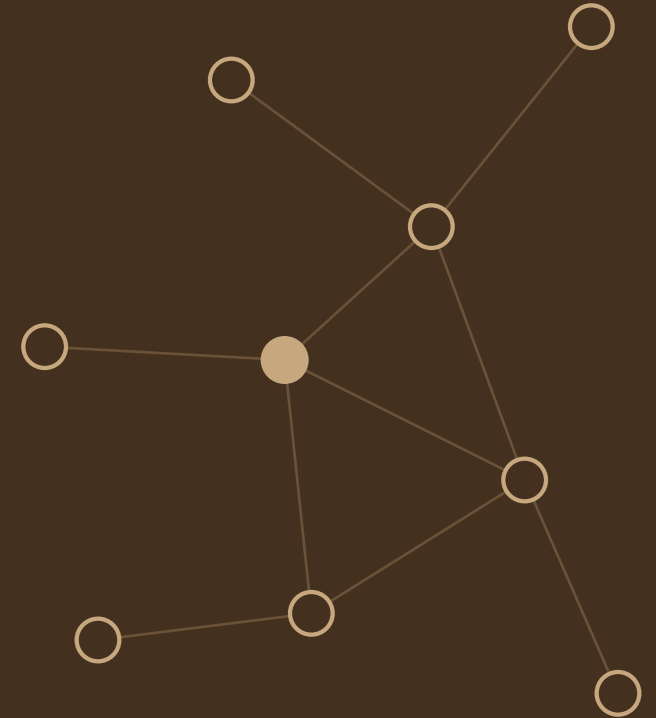
Adaptive lightweight cryptography that switches at runtime.

AUTHOR

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SUPERVISOR

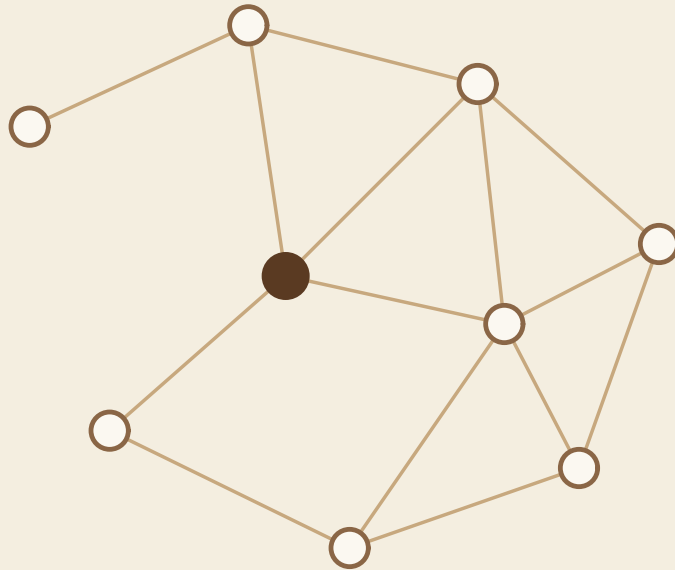
Oleg Bulichev



CONTEXT

Swarms are leaving the lab

— WHAT IT IS



agents linked by live, shifting wireless channels

— WHY IT MATTERS NOW

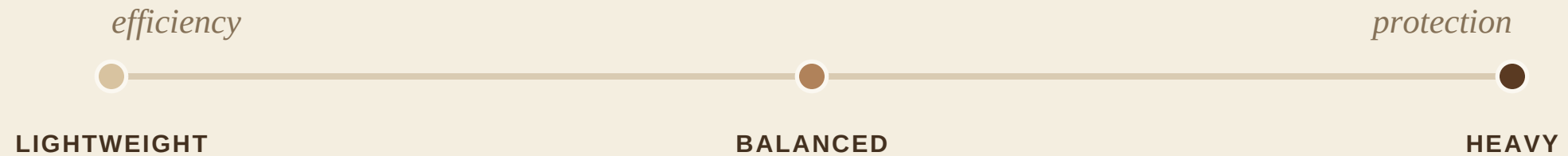
- **Communication is the system**
no central controller — agents act only on what they exchange
- **Already in the field**
search & rescue · agriculture · inspection · defence
- **An open research front**
lightweight crypto is active (NIST LWC, COSIC) — adapting it at runtime is not

Growing deployment makes protected swarm communication a timely, necessary problem.

THE PROBLEM

No single cipher fits a whole mission

Security and performance are coupled —
so balance them while the mission runs.



Always heavy

wastes throughput & battery

Always light

unsafe when threat rises

Strong primitives — but fixed for the whole mission

— WHAT THE FIELD OFFERS

- **Lightweight crypto**
ECC, AEAD, hash tokens — efficient on constrained nodes
- **Decentralised trust**
blockchain & attestation — strong assurance, heavy stack
- **Resilient comms**
self-healing relays, risk-aware routing — little built-in security

— THE GAP

- **Static configuration**
security parameters fixed for the whole mission
- **Fragmented layers**
trust, resilience & message protection studied apart
- **No executable comparison**
adaptive ideas rarely benchmarked against static baselines

Adaptive control of crypto settings is studied far less than the primitives themselves.

AIM & RESEARCH QUESTION

Does adapting crypto at runtime pay off?

— *can it hold the right protection at lower cost than a fixed profile?*

HOW THIS THESIS ANSWERS IT

01

MODEL

Swarm graph + adaptation function

$F : S(t) \rightarrow P$

02

PROTOTYPE

Pairwise testbed, four modes

incl. adaptive switching

03

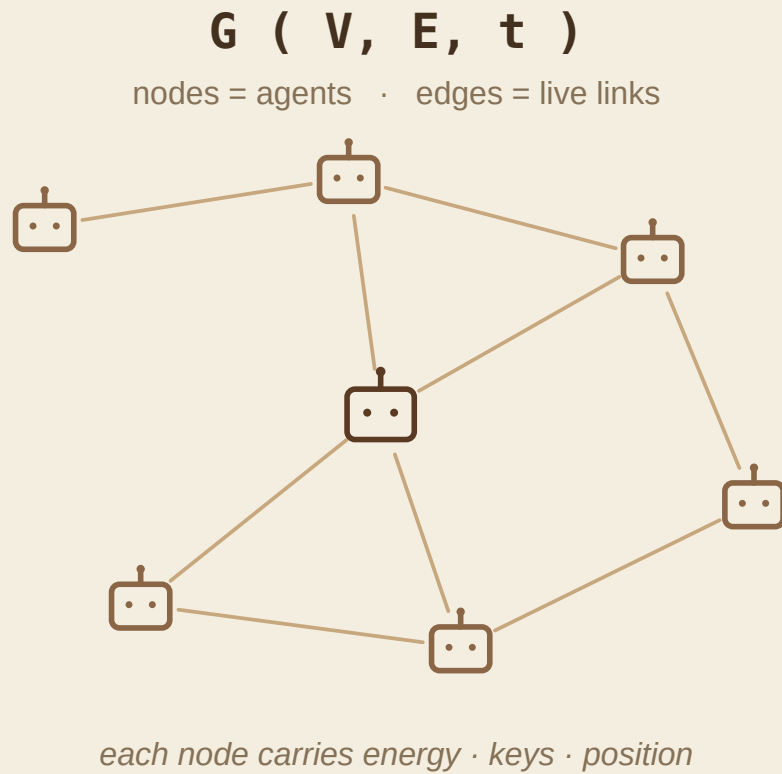
EVIDENCE

Measured in running code

228 benchmark runs

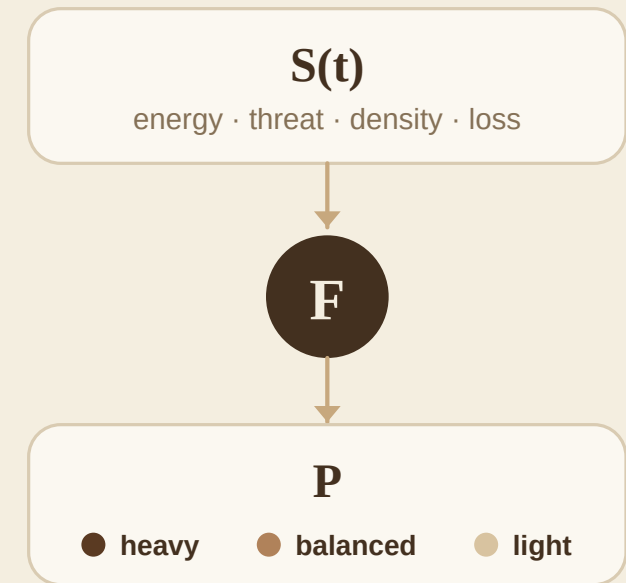
Novelty: a bounded formal model paired with an executable benchmark.

The swarm as a dynamic graph



— THE ADAPTATION FUNCTION

$$F : S(t) \rightarrow P$$



Three threats, and a data plane that adapts

— THE ADVERSARY CAN



Eavesdrop
read traffic



Tamper
inject / modify



Compromise
leak keys & memory

— OUR DEFENCE

SESSION SETUP

X25519 + HKDF-SHA256

forward secrecy · epoch rekeying

fixed

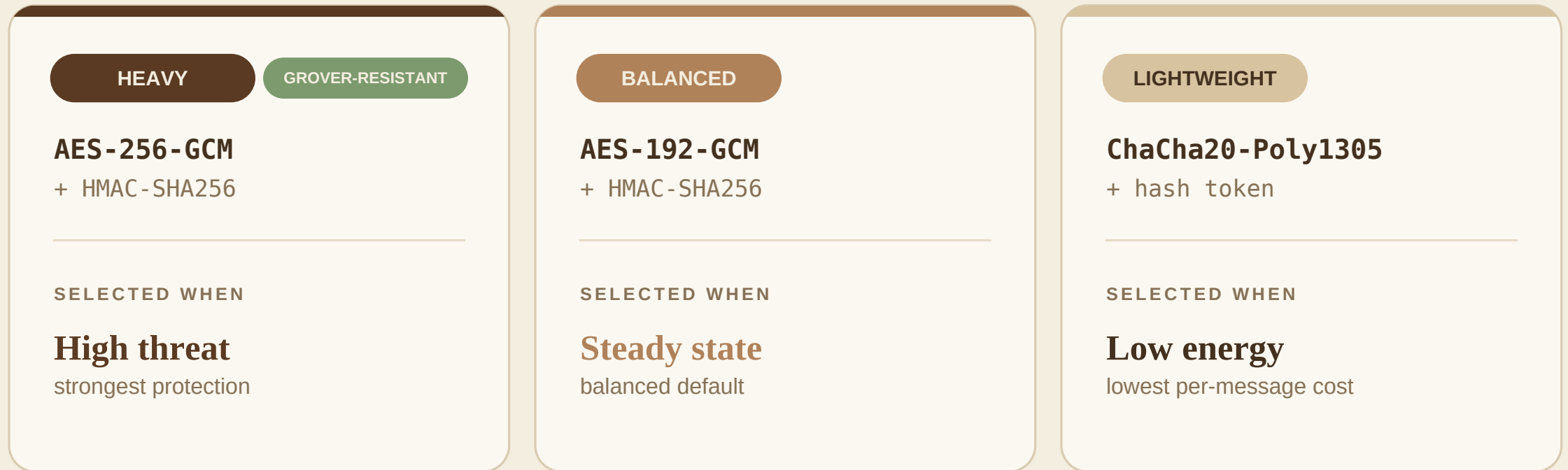


DATA PLANE

adaptive

- **Heavy** AES-256-GCM + HMAC
- **Balanced** AES-192-GCM + HMAC
- **Lightweight** ChaCha20-Poly1305 + token

Three operating points



Three operating points on one protection-versus-cost axis.

A simple rule moves between them at runtime: high threat selects heavy, low energy drops to lightweight, otherwise balanced.

BENCHMARK

A constrained, reproducible testbed



two containers · encrypted data path

0.05

CPU CORES

main · up to 0.25

128

MiB RAM

64

KiB / MSG

60

SECONDS

×3

REPEATS

Measured: latency · throughput · drops · switches · crypto work

Adaptive avoids both failure modes

Every static profile fails one way; adaptive fails neither.

PROFILE	UNDER-PROTECTS	OVER-PROTECTS	PROTECTION COST vs. always-heavy
Always-heavy	never	60% of run	100%
Always-balanced	40% of run	20% of run	67%
Always-light	80% of run	never	33%
Adaptive	never	never	73%

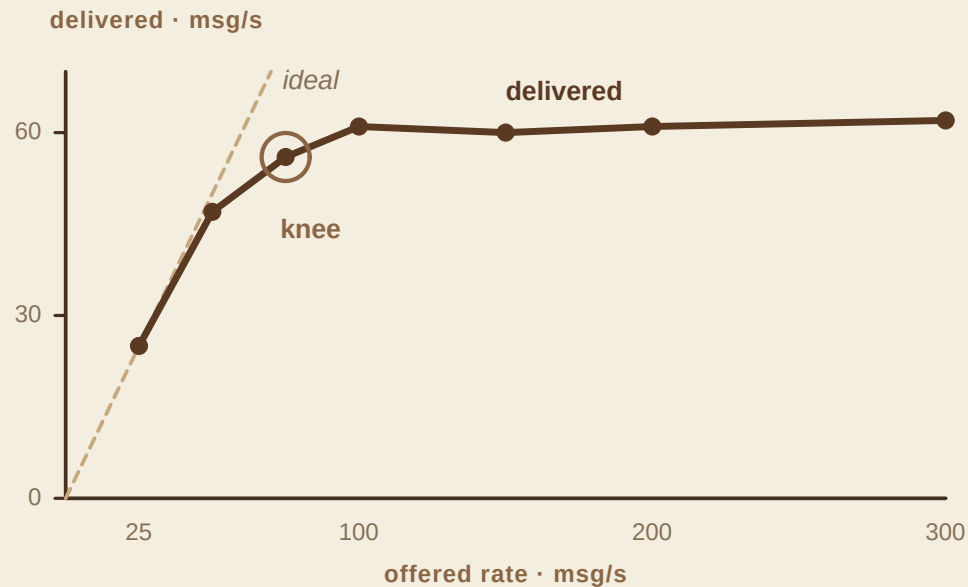
Only adaptive is never wrong either way — at 27% less protection work than always-heavy.

And the cost tracks the mission — a calmer schedule trends toward lightweight's cost, still escalating when it matters.

Under a tight CPU cap, transport dominates

DELIVERED vs OFFERED RATE

0.05-core cap



Near saturation the pipeline converges; the profile difference shows up in crypto work and policy fit.

CPU CAP REMOVES THE BOTTLENECK



The ceiling is CPU.

The crypto design keeps up.

CONCLUSION

Adaptive crypto, built and measured

✓ Formalised

swarm graph + $F : S(t) \rightarrow P$

✓ Built

four modes incl. adaptive switching

✓ Measured

lightweight cheapest; adaptive allocates well

WHAT COMES NEXT

01 Authenticated control plane

signed bootstrap & updates

02 Multi-peer validation

churn · contention · async

03 Hardware & post-quantum

real platforms; hybrid PQ

Thank you — questions welcome.

github.com/mazzz3r/adaptive-swarm-crypto-thesis

Limitations & scope

- **Pairwise prototype**

two peers on the measured path; multi-hop & churn are out of scope

- **Energy is a model proxy**

valid for relative comparison only, not battery-accurate

- **Data plane only**

no authenticated control plane (signed bootstrap / updates)

- **Host environment**

Docker + gRPC containers; no embedded radios or microcontrollers

- **Scripted inputs**

threat & energy are scripted, not sensed or estimated

- **Coarse cost metric**

crypto-work counts bytes, so heavy $\approx$ balanced; thresholds set by hand

Every claim covers exactly what the constrained pairwise prototype actually runs.